Supplementary Material

Soil-Water Modelling, Decision Support and a Graphical User Interface for the Daisy Model

Derivations of the equations, the mechanisms behind the results, the data, and the origin of every table and figure

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S1. What is new in this article

The table sets each contribution against what was established before it. The derivations that follow mark which equations are taken from the literature and which are new.

Table S1: What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

#	Established before	New in this article	Evidence here
1	Daisy is run from text set-up files (Hansen et al., 2012); field, design, satellite and model tools are separate programs.	A graphical workbench that checks every design against a catalogue of what Daisy can vary, keeps an exact run budget, keeps every run and explains every failure by its root cause.	End-to-end Daisy 7.1.14 runs of the library scenarios, deficit irrigation and the soil-hydraulic factors; a defect that stopped every run on a sample horizon found and removed.
2	Decision tools show numbers without their origin.	Two auditable rules: every number leads to an equation with its origin, and a Daisy layer shows a Daisy result only from a Daisy run.	All layers opened without a page error; Daisy layers show nothing before a run.
3	Shared model functions are trusted once published.	A check of a shared function against stored Daisy results that found a defect.	The denitrification function corrected after comparison with 792 Daisy results.
4	Sentinel-2 indices, SoilGrids and elevation models are used separately (Drusch et al., 2012; Poggio et al., 2021).	One offline chain from a field outlined on a satellite image to indices, water sources, terrain, soil loss, retention maps, a design and a schedule.	Run on a real 5.85 ha field near Zaria, Nigeria.
5	Sensitivity to hydraulic parameters is studied with dedicated codes (Šimůnek et al., 2016).	Factors that scale the van Genuchten–Mualem parameters of every horizon of a Daisy run inside the workbench.	One season's matrix percolation rose from 177 to 222 mm and runoff vanished with the scaled parameters.
6	–	A manual made with the program and opened at the page in use (Halilu, 2026).	Every control, view and equation documented; page links from each screen.

S2. The author's contributions: problem, derivation and pinpoints

Each contribution of the article is set out in the same way: *the problem* it answers, stated in a few lines; *how the author developed it*; *the result*, with the parts that matter marked by circled labels (1), (2), …; and *the pinpoints*, which say for each circled mark what it shows and why it is so. Every equation that the author developed is circled in red pencil and labelled *developed by the author*. Numbers come from the data and can be recomputed in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`.

S2.1 Contribution 1: the rule of origin

The problem

Decision tools that couple a process model with fast approximations can show a number that looks like a model result but was not produced by the model. How can a user be sure that a "Daisy" number came from Daisy?

How the author developed it

Every value passed by the bridge carries the log file and line it was read from; before drawing, a layer tests that origin and draws nothing if there is no run, stating which run is needed.

The result

$$\underbrace{\text{show}(v)}_{(1)} \iff \underbrace{\text{source}(v) \in \{\text{Daisy runs of this project}\}}_{(2)}$$

$$\underbrace{\text{no run} \Rightarrow \text{empty layer with the reason}}_{(3)}$$

developed by the author

Pinpoints: what each circled mark shows and why

(1) A value of a Daisy layer is drawn only if the condition on the right holds.

(2) The source of every value is the log file and line of a run; the bridge carries it to the layer.

(3) Before any run the layer says what it needs and shows no example numbers, which was checked for every layer. The rule makes the origin auditable instead of trusted.

S2.2 Contribution 2: an exact run budget that still tests every level

The problem

A design of a few factors can need hundreds of Daisy runs of minutes each. How can the user set a budget and still test every level of every factor?

How the author developed it

The full factorial is counted; if it exceeds the budget B, a Latin hypercube of B runs is drawn in which the levels of each factor are used as evenly as possible.

The result

$$\underbrace{N}_{(1)} = \min\Big(\underbrace{\textstyle\prod_i n_i}_{(2)},\ \underbrace{B}_{(3)}\Big)$$

$$\underbrace{\text{each level of factor } i \text{ used } \lfloor N/n_i \rfloor \text{ or } \lceil N/n_i \rceil \text{ times}}_{(4)}$$

developed by the author

Pinpoints: what each circled mark shows and why

(1) The number of runs actually made, never above the budget.

(2) The full factorial (for 4, 10 and 5 levels: 200 runs).

(3) The budget set by the user (for example 60).

(4) The Latin-hypercube choice spreads the runs so that every level appears almost equally often (each of the 10 levels 6 times), so main effects remain estimable.

S2.3 Contribution 3: the root cause of a failed run

The result

Daisy message contains	Class	What to do
(1) Unknown '…' model	a name not defined	take only the needed definitions
(2) did not converge, NaN	numerical	bring extreme values closer to typical
(3) out of range, must be	range	change the level that writes the value
(4) cannot open, not found	missing file	rerun; check the weather and soil

Pinpoints: what each circled mark shows and why

(1) This class found a real defect: a file of the Daisy samples referred to logs of its own run; taking only the definitions a run needs removed it.

(2) Numerical failures come from extreme hydraulic values; the message names the factor and the typical range.

(3) The line and column of the set-up file are shown, so the user sees which input wrote the value.

(4) Failed runs are grouped by cause, so one explanation covers all runs that failed for the same reason.

S2.4 Contribution 4: soil-hydraulic factors for every horizon

How the author developed it

Each factor rewrites the hydraulic block of every horizon of the set-up file before Daisy runs; $n-1$ rather than n is scaled so that $n>1$ for any factor; one run of the loam was made with Daisy 7.1.14 with and without the factors.

The result

$\underbrace{K_s \to f_k K_s}_{(1)},\quad \underbrace{\alpha \to f_a \alpha}_{(2)},\quad \underbrace{n-1 \to f_n(n-1)}_{(3)},\quad \theta_s \to f_t \theta_s,\quad \ell \to \ell+\delta\ell;\quad \underbrace{\text{percolation } 177 \to 2}_{(4)}$

developed by the author

Pinpoints: what each circled mark shows and why

(1) Scaling the saturated conductivity of every horizon at once keeps the layering of the soil.

(2) Scaling α moves the air-entry suction.

(3) Scaling $n - 1$ (not n) keeps $n > 1$, so the van Genuchten curve stays valid for any factor.

(4) A faster-draining parameter set, run end to end with Daisy, raised one season's matrix percolation from 177 to 222 mm and removed runoff, the direction expected from the conductivity.

S2.5 Contribution 5: a defect found in a shared function

The problem

The denitrification function is shared with Daisy's Python tools. Is the copy used by the software right?

How the author developed it

The function was evaluated on the stored inputs of the 792 cases distributed with Daisy's tools and its outputs compared case by case with Daisy's stored results.

The result

$$\underbrace{\max_{j=1..792} \left| D_j^{\text{SWAFL}} - D_j^{\text{Daisy}} \right|}_{(1)} \quad \Rightarrow \quad \underbrace{\text{one branch wrong, corrected}}_{(2)}$$

developed by the author

Pinpoints: what each circled mark shows and why

(1) The function was evaluated on the 792 cases for which Daisy's own results are stored, and compared case by case.

(2) One class of cases disagreed; the branch responsible was found and corrected, after which the cases agree. A shared function was tested rather than trusted.

S3. A guide to the results with worked numbers

Each part explains one computation of the software with a worked example; the numbers are in sheet `EQ_P3` of the workbook.

S3.1 1. a vegetation index from a satellite pixel

Green leaves reflect near-infrared light and absorb red light. *Example:* near-infrared reflectance 0.28 and red 0.11 give NDVI $= (0.28 - 0.11)/(0.28 + 0.11) = 0.44$, a moderately green crop.

S3.2 2. the crop coefficient of the day

FAO-56 describes the crop's water use relative to a reference grass with four straight lines. *Example:* with stages of 30, 40, 50 and 30 days and K_c of 0.40, 1.15 and 0.40, day 50 lies in the development stage: $K_c = 0.40 + (1.15 - 0.40) \times (50 - 30)/40 = 0.78$; day 70 starts the mid-season, $K_c = 1.15$.

S3.3 3. the daily water balance of a map cell

Each 10 m cell keeps a root-zone depletion: yesterday's depletion minus rain and irrigation plus crop water use; when it passes the readily available water, the cell needs irrigation of that depth.

S3.4 4. soil loss

$A = R\,K\,LS\,C\,P$. *Example:* $450 \times 0.03 \times 0.4 \times 0.3 \times 1 = 1.6$ t ha^{-1} yr^{-1}, below a tolerable 2 t.

S3.5 5. where the soil stays wet

The wetness index is $\ln(a/\tan\beta)$. *Example:* 250 m of upslope area per metre of contour on a slope of 1.9° gives $\ln(250/0.0332) = 8.9$, a wet position.

S3.6 6. keeping an experiment within its budget

Three factors with 4, 10 and 5 levels make 200 combinations; with a budget of 60 runs a Latin-hypercube choice makes 60 runs in which each level of the second factor appears 6 times, so every level is still tested.

S3.7 7. splitting denitrification

With $12\,\mathrm{kg\,N\,ha^{-1}}$ denitrified and an N_2:N_2O ratio of 4, $N_2O = 12/5 = 2.4$ and $N_2 = 9.6\,\mathrm{kg\,N\,ha^{-1}}$.

S3.8 8. the rule of origin

A decision layer may show a Daisy result only if a Daisy run made it. Before a run, the layer therefore says that it needs a run and shows nothing; after a run it shows the run's numbers with their file and line.

S4. Derivations of the equations

The derivations of Section S2 are of established methods (spectral indices, FAO-56, the Richards equation, budgets, D8 flow and wetness, RUSLE, Latin-hypercube sampling), given so that each computation of the software can be followed. They are not claimed as new; the contribution of the article is the integration, the two rules of origin, and the verification that found defects (Table S1).

S4.1 Normalised spectral indices

Green leaves absorb red light for photosynthesis and scatter near-infrared light in the leaf mesophyll. The normalised difference NDVI $= (\rho_N - \rho_R)/(\rho_N + \rho_R)$ is a contrast of the two that is bounded in $[-1, 1]$ and cancels a common multiplicative illumination factor k: $(k\rho_N - k\rho_R)/(k\rho_N + k\rho_R)$ equals NDVI. SAVI adds a soil term L to the denominator to reduce the influence of bright soil, $(1 + L)(\rho_N - \rho_R)/(\rho_N + \rho_R + L)$; EVI corrects for the atmosphere with the blue band and for the canopy background; the water indices use the shortwave-infrared band, which liquid water absorbs.

S4.2 Crop coefficient through the season

The FAO-56 curve is piecewise linear in time: constant K_{ini} during the initial stage L_1, a linear rise to K_{mid} over the development stage L_2, constant over the mid-season L_3 and a linear fall to K_{end} over the late stage L_4 (Allen et al., 1998). The crop evapotranspiration is $ET_c = K_c\,ET_0$.

S4.3 Daily root-zone balance of each map cell

For each cell, the root-zone depletion follows $D_{r,i} = D_{r,i-1} - (P - \mathrm{RO})_i - I_i + ET_{c,i} + \mathrm{DP}_i$, bounded by $0 \le D_r \le TAW$; deep percolation occurs only when the depletion would fall below zero. Irrigation is required when D_r exceeds the readily available water $p\,TAW$; the net depth to apply is D_r (Allen et al., 1998).

S4.4 The Richards equation of the soil column

Mass conservation of water in a layer of thickness dz gives $\partial\theta/\partial t = -\partial q/\partial z - S$, with S the root uptake. The Darcy–Buckingham flux with z positive downward and h the suction head is $q = K(h)(1 - \partial h/\partial z)$: gravity drives water down and a suction gradient drives it towards the drier soil. With $\theta(h)$ and $K(h)$ from the van Genuchten–Mualem functions the equation is closed and is solved implicitly with Picard iteration in the column of the workbench.

S4.5 Nitrogen and chemical budgets

For each season, the change of the mineral nitrogen store equals inputs minus outputs: fertiliser, deposition and mineralisation in; uptake, leaching, denitrification and volatilisation out. For a chemical, $\Delta M = A - U - L - R - X$ (applied, taken up, leached by every route, run off, transformed); the bridge reads each term from the Daisy logs and the closure of this budget is its test.

S4.6 Splitting denitrification into N_2O and N_2

Let r be the ratio of N_2 to N_2O in the gaseous product of denitrification D. Then $D_{N_2O} + D_{N_2} = D$ and $D_{N_2} = r\,D_{N_2O}$, so $D_{N_2O}(1 + r) = D$, giving $D_{N_2O} = D/(1 + r)$ and $D_{N_2} = rD/(1 + r) = D/(1 + 1/r)$, the two expressions of the paper. The ratio is the smaller of a nitrate and a carbon-dioxide limited ratio, scaled by a water factor, so wetter, more active soils give relatively more N_2.

S4.7 Terrain, flow and wetness

On a filled elevation grid each cell drains to the steepest of its eight neighbours (D8). The upslope area A of a cell is the sum of the areas that drain through it, and the specific area is $a = A/w$ with

w the cell width. The topographic wetness index $\ln(a/\tan\beta)$ is large where much water arrives and the slope β is small, which marks where the soil stays wet.

S4.8 Soil loss

The revised universal soil loss equation multiplies the rain erosivity R, the soil erodibility K, the slope length and steepness LS, the cover C and the practice P: $A = R\,K\,LS\,C\,P$ in t ha^{-1} yr^{-1} (Renard and Freimund, 1994).

S4.9 Run budget of an experiment

For k factors with $n_1, \ldots, n_k$ levels the full design has $\prod n_i$ runs. When this exceeds the budget B set by the user, a Latin-hypercube choice of B runs is made: each factor's levels are used as evenly as possible (B/n_i times each), so every level of every factor appears and the main effects remain estimable while the number of runs never exceeds B.

S5. Why the results happen

Why every number can be traced. Each value shown is computed by a function that carries the number of its equation and a label of origin (literature, fitted to data, engineered); the interface shows both, so the origin is a property of the code, not of the documentation.

Why a Daisy layer draws nothing before a Daisy run. The rule of origin forbids a layer to show a Daisy result from any other source; without a run there is no result, so the layer says so rather than showing an example.

Why the compact designs lose some shape of a curve. A design with fewer levels samples a curve at fewer points; the marginal curve is then drawn through those points and misses curvature between them, which is the loss measured.

Why the soil column is slow for a ponded fine soil. With ponding the top boundary is a fixed head and the fine soil has a very small conductivity and a steep head gradient at the surface; the implicit solver then needs small time steps and many iterations to converge.

Why the denitrification defect was found. The check compared the shared function with the stored Daisy results over 792 cases; one case class disagreed, which pointed to a wrong branch in the

function, corrected in the release.

Why runs on a horizon of the Daisy samples failed. The sample file also defined logs that referred to a model of its original run; Daisy stopped on the unknown name. Taking only the definitions a run needs removed the error.

Why the routes from Daisy to a decision differ in time by orders of magnitude. Running Daisy solves the Richards equation and a crop for every day of every run; the meta-model looks up and combines stored outcomes, which takes microseconds.

S6. The data

Table S2 lists the 17 experiments of the data folder and Fig. S1 shows their outputs. The tables that follow are the result files of the analyses of the paper, printed in full or up to 220 rows; each is also in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`, and every run is in `Data/Comprehensive_Summary`.

Table S2: The data of this study: the 17 Daisy experiments of `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; Y yield (Mg ha^{-1}), I irrigation, D drainage at 1 m and ET (mm yr^{-1}), N_L nitrate leached (kg N ha^{-1} yr^{-1}), t run time of Daisy (s). Every row is one file of the folder, linked.

Exp.	File	Runs	Valid	Factors varied	Y	I	D	ET	N_L	t
E0	`Baselines`	10	10	soil	2.19	0	55	418	158	231
E1	`Doses`	200	189	soil, rate, dose	5.13	198	98	564	79	195
E2	`TriggerPressure`	200	197	soil, trigger	4.70	169	80	559	90	139
E3	`SensorDepth`	200	200	soil, sensor depth	4.94	199	86	583	86	146
E4	`Methods`	100	96	soil, method	5.01	164	88	542	83	147
E5	`SHP Dynamics`	300	300	soil, method, tillage	4.13	152	43	579	84	189
E6	`Mediterranean Pollution`	210	193	soil, method, trigger, N rate	3.64	84	239	620	62	118
E7	`Mulching`	200	200	soil, mulch	5.29	183	96	555	74	161
E8	`Rain Intensity Moisture`	1000	1000	initial suction, rain scale	0.68	0	-1	282	-0	6
E9	`Soil Moisture Rain`	1000	987	soil, initial suction, rain scale	0.75	0	19	247	33	7
E10	`Amount Timing Soil`	1000	954	soil, rate, dose, window	4.42	139	76	532	97	146
E11	`Temp Evap Rain`	1000	1000	rain scale, warming, demand	3.81	324	18	754	66	151
E12	`Sensor Trigger Rain`	1000	1000	trigger, sensor depth, rain scale	5.88	260	35	670	32	155
E13	`N Water Soil`	1000	975	soil, method, rate, dose, trigger, N rate	4.00	162	83	547	79	207
E14	`Mulch Rain`	1000	1000	rain scale, mulch	6.15	235	43	634	25	173
E15	`Method Depth Amount`	1000	1000	method, rate, dose, root depth	6.02	217	61	619	37	157
E16	`Edge Cases`	770	740	soil, method, rate, dose, trigger, sensor depth, rain scale, warming, N rate, demand	2.17	344	185	600	93	224

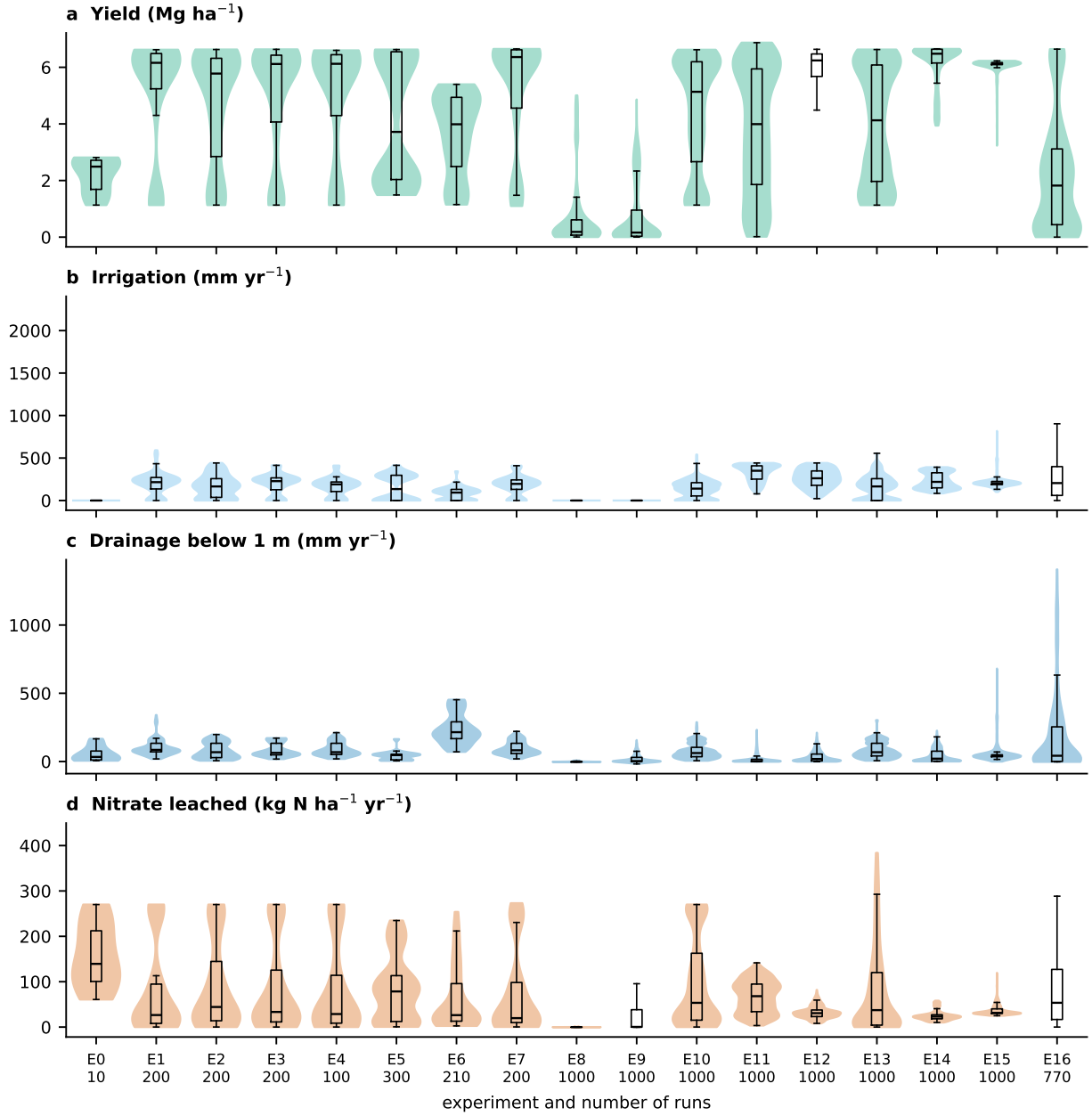


Figure S1: Yield, irrigation, drainage and nitrate leaching of the valid runs, by experiment.

out_p4_calibration.json

Source: `Manuscripts/reproducibility/p456/out_p4_calibration.json`; 140 values.

quantity	value
truth.alphaF	0.05303
truth.Kd	0.02083
truth.alpha	0.0001
loss0	1.409e-31
start	0.204
runs[0].method	sequential
runs[0].seed	1

quantity	value
runs[0].evals	16
runs[0].loss	0.0006235
runs[0].x.alphaF	0.05
runs[0].x.Kd	0.06
runs[0].x.alpha	0.0003
runs[0].ms	20
runs[0].err	0.02553, 0.4594, 0.4771
runs[1].method	sequential
runs[1].seed	2
runs[1].evals	16
runs[1].loss	0.0006235
runs[1].x.alphaF	0.05
runs[1].x.Kd	0.06
runs[1].x.alpha	0.0003
runs[1].ms	4
runs[1].err	0.02553, 0.4594, 0.4771
runs[2].method	sequential
runs[2].seed	3
runs[2].evals	16
runs[2].loss	0.0006235
runs[2].x.alphaF	0.05
runs[2].x.Kd	0.06
runs[2].x.alpha	0.0003
runs[2].ms	4
runs[2].err	0.02553, 0.4594, 0.4771
runs[3].method	sequential
runs[3].seed	4
runs[3].evals	16
runs[3].loss	0.0006235
runs[3].x.alphaF	0.05
runs[3].x.Kd	0.06
runs[3].x.alpha	0.0003
runs[3].ms	4
runs[3].err	0.02553, 0.4594, 0.4771
runs[4].method	sequential
runs[4].seed	5
runs[4].evals	16
runs[4].loss	0.0006235
runs[4].x.alphaF	0.05
runs[4].x.Kd	0.06
runs[4].x.alpha	0.0003
runs[4].ms	4
runs[4].err	0.02553, 0.4594, 0.4771
runs[5].method	cma
runs[5].seed	1
runs[5].evals	154
runs[5].loss	4.312e-06
runs[5].x.alphaF	0.0533

quantity	value
runs[5].x.Kd	0.004069
runs[5].x.alpha	2.777e-05
runs[5].ms	39
runs[5].err	0.002251, 0.7093, 0.5564
runs[6].method	cma
runs[6].seed	2
runs[6].evals	154
runs[6].loss	4.144e-06
runs[6].x.alphaF	0.05313
runs[6].x.Kd	0.006077
runs[6].x.alpha	0.000431
runs[6].ms	36
runs[6].err	0.0008651, 0.535, 0.6345
runs[7].method	cma
runs[7].seed	3
runs[7].evals	154
runs[7].loss	2.003e-06
runs[7].x.alphaF	0.05286
runs[7].x.Kd	0.05183
runs[7].x.alpha	9.474e-05
runs[7].ms	38
runs[7].err	0.001402, 0.3958, 0.02349
runs[8].method	cma
runs[8].seed	4
runs[8].evals	154
runs[8].loss	6.869e-06
runs[8].x.alphaF	0.05337
runs[8].x.Kd	0.02493
runs[8].x.alpha	2.533e-05
runs[8].ms	36
runs[8].err	0.002804, 0.07801, 0.5964
runs[9].method	cma
runs[9].seed	5
runs[9].evals	154
runs[9].loss	9.754e-09
runs[9].x.alphaF	0.05301
runs[9].x.Kd	0.003502
runs[9].x.alpha	0.0001228
runs[9].ms	37
runs[9].err	0.0001234, 0.7744, 0.08912
runs[10].method	bayes
runs[10].seed	1
runs[10].evals	150
runs[10].loss	6.888e-08
runs[10].x.alphaF	0.05305
runs[10].x.Kd	0.04165
runs[10].x.alpha	2.105e-05
runs[10].ms	1779

quantity	value
runs[10].err	0.000177, 0.3008, 0.6768
runs[11].method	bayes
runs[11].seed	2
runs[11].evals	150
runs[11].loss	7.706e-08
runs[11].x.alphaF	0.05307
runs[11].x.Kd	0.09905
runs[11].x.alpha	6.412e-05
runs[11].ms	1671
runs[11].err	0.0003446, 0.6771, 0.193
runs[12].method	bayes
runs[12].seed	3
runs[12].evals	150
runs[12].loss	8.24e-08
runs[12].x.alphaF	0.05306
runs[12].x.Kd	0.009166
runs[12].x.alpha	1.795e-05
runs[12].ms	1638
runs[12].err	0.0003083, 0.3566, 0.746
runs[13].method	bayes
runs[13].seed	4
runs[13].evals	150
runs[13].loss	1.321e-07
runs[13].x.alphaF	0.05302
runs[13].x.Kd	0.003837
runs[13].x.alpha	2.652e-05
runs[13].ms	1586
runs[13].err	7.897e-05, 0.7348, 0.5764
runs[14].method	bayes
runs[14].seed	5
runs[14].evals	150
runs[14].loss	2.571e-07
runs[14].x.alphaF	0.053
runs[14].x.Kd	0.05913
runs[14].x.alpha	1.386e-05
runs[14].ms	1666
runs[14].err	0.000195, 0.453, 0.8582

out_p4_design.json

Source: `Manuscripts/reproducibility/p456/out_p4_design.json`; 96 values.

quantity	value
[0].id	E0
[0].name	Baselines
[0].gui	9

quantity	value
[0].study	10
[0].guiFactorCombos	9
[0].ensCombos	0
[1].id	E1
[1].name	Dose and duration
[1].gui	17
[1].study	200
[1].guiFactorCombos	16
[1].ensCombos	1
[2].id	E2
[2].name	Trigger pressure
[2].gui	21
[2].study	200
[2].guiFactorCombos	18
[2].ensCombos	200
[3].id	E3
[3].name	Sensor depth
[3].gui	21
[3].study	200
[3].guiFactorCombos	18
[3].ensCombos	200
[4].id	E4
[4].name	Methods
[4].gui	6
[4].study	100
[4].guiFactorCombos	4
[4].ensCombos	10
[5].id	E5
[5].name	Soil structure dynamics (SHP)
[5].gui	72
[5].study	300
[5].guiFactorCombos	72
[5].ensCombos	10
[6].id	E6
[6].name	Mediterranean pollution
[6].gui	30
[6].study	210
[6].guiFactorCombos	30
[6].ensCombos	10
[7].id	E7
[7].name	Mulching
[7].unsupported	True
[7].study	200
[8].id	E8
[8].name	Rain intensity and moisture
[8].unsupported	True
[8].study	1000
[9].id	E9

quantity	value
[9].name	Soil moisture and rain
[9].gui	40
[9].study	1000
[9].guiFactorCombos	32
[9].ensCombos	1000
[10].id	E10
[10].name	Amount, timing and soil
[10].gui	20
[10].study	1000
[10].guiFactorCombos	18
[10].ensCombos	100
[11].id	E11
[11].name	Temperature, evaporation, rain
[11].gui	16
[11].study	1000
[11].guiFactorCombos	16
[11].ensCombos	100
[12].id	E12
[12].name	Sensor, trigger, rain
[12].gui	30
[12].study	1000
[12].guiFactorCombos	27
[12].ensCombos	1000
[13].id	E13
[13].name	N, water, soil
[13].gui	32
[13].study	1000
[13].guiFactorCombos	24
[13].ensCombos	10
[14].id	E14
[14].name	Mulch and rain
[14].unsupported	True
[14].study	1000
[15].id	E15
[15].name	Method, depth, amount
[15].gui	30
[15].study	1000
[15].guiFactorCombos	27
[15].ensCombos	100
[16].id	E16
[16].name	Edge cases
[16].gui	24
[16].study	770
[16].guiFactorCombos	16
[16].ensCombos	5

out_p4_fidelity.csv

Source: `Manuscripts/reproducibility/p456/out_p4_fidelity.csv`; 51 rows. Workbook sheet (`CSV file`).

exp	factor	resp	levels_study	levels_compact	rmse_pct	max_pct
E2	trig	yield_Mg_ha	20	6	7.317	26.9
E2	trig	irrigation_mm	20	6	6.443	22.11
E2	trig	drainage_1m_mm	20	6	8.957	36.94
E2	trig	IE_kg_m3	20	6	8.133	28.55
E2	trig	N_leached_kg_ha	20	6	7.183	23.64
E3	z	yield_Mg_ha	20	6	4.638	11.63
E3	z	irrigation_mm	20	6	11.23	24.55
E3	z	drainage_1m_mm	20	6	4.563	9.679
E3	z	IE_kg_m3	20	6	6.511	13.93
E3	z	N_leached_kg_ha	20	6	4.849	13
E9	trig	yield_Mg_ha	10	4	11.68	33.78
E9	trig	drainage_1m_mm	10	4	26.19	79.45
E9	trig	N_leached_kg_ha	10	4	16.2	49.6
E9	P	yield_Mg_ha	10	4	10.03	27.69
E9	P	drainage_1m_mm	10	4	12.32	35
E9	P	N_leached_kg_ha	10	4	7.935	20.58
E10	amt_mm	yield_Mg_ha	10	3	32.48	96.29
E10	amt_mm	irrigation_mm	10	3	29.85	88.58
E10	amt_mm	drainage_1m_mm	10	3	36.08	88.19
E10	amt_mm	IE_kg_m3	10	3	27.91	85.16
E10	amt_mm	N_leached_kg_ha	10	3	30.38	90.76
E11	dT	yield_Mg_ha	10	4	8.061	19.75
E11	dT	irrigation_mm	10	4	11.16	29.98
E11	dT	drainage_1m_mm	10	4	14.95	42.6
E11	dT	IE_kg_m3	10	4	10.98	30.61
E11	dT	N_leached_kg_ha	10	4	13.5	34.91
E11	P	yield_Mg_ha	10	4	6.598	14.63
E11	P	irrigation_mm	10	4	9.053	24.73
E11	P	drainage_1m_mm	10	4	16.9	48.6
E11	P	IE_kg_m3	10	4	10.36	28.78
E11	P	N_leached_kg_ha	10	4	9.729	21.93
E12	z	yield_Mg_ha	10	3	11.16	23.65
E12	z	irrigation_mm	10	3	13.24	24.2
E12	z	drainage_1m_mm	10	3	27.73	65.95
E12	z	IE_kg_m3	10	3	21.02	50.32
E12	z	N_leached_kg_ha	10	3	29.31	64.31
E12	trig	yield_Mg_ha	10	3	15.58	42.43
E12	trig	irrigation_mm	10	3	13.2	28.82
E12	trig	drainage_1m_mm	10	3	17.55	52.43
E12	trig	IE_kg_m3	10	3	18.63	54.2
E12	trig	N_leached_kg_ha	10	3	16.13	44.94
E12	P	yield_Mg_ha	10	3	21.89	64.7

exp	factor	resp	levels_study	levels_compact	rmse_pct	max_pct
E12	P	irrigation_mm	10	3	6.649	16.43
E12	P	drainage_1m_mm	10	3	8.067	23.93
E12	P	IE_kg_m3	10	3	10.71	31.48
E12	P	N_leached_kg_ha	10	3	30.16	89.24
E15	root	yield_Mg_ha	10	3	18.15	56.06
E15	root	irrigation_mm	10	3	31.64	100
E15	root	drainage_1m_mm	10	3	27.96	88.29
E15	root	IE_kg_m3	10	3	15.45	47.71
E15	root	N_leached_kg_ha	10	3	14.68	44.85

out_p4_richards_js.json

Source: `Manuscripts/reproducibility/p456/out_p4_richards_js.json`; 153 values.

quantity	value
SOILS.sand.alpha	0.145
SOILS.sand.n	2.68
SOILS.sand.Ks	29.7
SOILS.sand.Ths	0.43
SOILS.sand.Thr	0.045
SOILS.sand.l	0.5
SOILS.loam.alpha	0.036
SOILS.loam.n	1.56
SOILS.loam.Ks	1.04
SOILS.loam.Ths	0.43
SOILS.loam.Thr	0.078
SOILS.loam.l	0.5
SOILS.siltclayloam.alpha	0.01
SOILS.siltclayloam.n	1.23
SOILS.siltclayloam.Ks	0.07
SOILS.siltclayloam.Ths	0.5
SOILS.siltclayloam.Thr	0.089
SOILS.siltclayloam.l	0.5
init.sand	45.68
init.loam	170.1
init.siltclayloam	394.8
out.sand.1.infil	200
out.sand.1.drain	152.1
out.sand.1.runoff	0
out.sand.1.store0	None
out.sand.1.storeEnd	93.56
out.sand.1.maxErr	0
out.sand.1.sumErr	0
out.sand.1.iters	9964
out.sand.1.ms	1586
out.sand.1.infilD	200, 0, 0, 0, 0, 0

quantity	value
out.sand.1.drainD	18.79, 96.26, 18.37, 8.962, 5.692, 4.046
out.sand.1.store	226.9, 130.6, 112.3, 103.3, 97.6, 93.56
out.sand.2.infil	200
out.sand.2.drain	152.1
out.sand.2.runoff	0
out.sand.2.store0	None
out.sand.2.storeEnd	93.56
out.sand.2.maxErr	0
out.sand.2.sumErr	0
out.sand.2.iters	6306
out.sand.2.ms	654
out.sand.2.infilD	200, 0, 0, 0, 0, 0
out.sand.2.drainD	18.79, 96.23, 18.37, 8.982, 5.7, 4.051
out.sand.2.store	226.9, 130.7, 112.3, 103.3, 97.61, 93.56
out.sand.4.infil	200
out.sand.4.drain	152.2
out.sand.4.runoff	0
out.sand.4.store0	None
out.sand.4.storeEnd	93.53
out.sand.4.maxErr	0
out.sand.4.sumErr	0
out.sand.4.iters	4192
out.sand.4.ms	298
out.sand.4.infilD	200, 0, 0, 0, 0, 0
out.sand.4.drainD	18.79, 96.16, 18.45, 9.001, 5.703, 4.054
out.sand.4.store	226.9, 130.7, 112.3, 103.3, 97.58, 93.53
out.sand.0.5.infil	200
out.sand.0.5.drain	152.1
out.sand.0.5.runoff	0
out.sand.0.5.store0	None
out.sand.0.5.storeEnd	93.54
out.sand.0.5.maxErr	0
out.sand.0.5.sumErr	0
out.sand.0.5.iters	15784
out.sand.0.5.ms	4958
out.sand.0.5.infilD	200, 0, 0, 0, 0, 0
out.sand.0.5.drainD	18.79, 96.27, 18.37, 8.99, 5.676, 4.047
out.sand.0.5.store	226.9, 130.6, 112.2, 103.3, 97.59, 93.54
out.loam.1.infil	200
out.loam.1.drain	54.59
out.loam.1.runoff	0
out.loam.1.store0	None
out.loam.1.storeEnd	315.5
out.loam.1.maxErr	0
out.loam.1.sumErr	0
out.loam.1.iters	10486
out.loam.1.ms	1599
out.loam.1.infilD	200, 0, 0, 0, 0, 0

quantity	value
out.loam.1.drainD	0.009, 5.772, 19.08, 12.88, 9.458, 7.396
out.loam.1.store	370.1, 364.3, 345.2, 332.3, 322.9, 315.5
out.loam.2.infil	200
out.loam.2.drain	54.63
out.loam.2.runoff	0
out.loam.2.store0	None
out.loam.2.storeEnd	315.4
out.loam.2.maxErr	0
out.loam.2.sumErr	0
out.loam.2.iters	8549
out.loam.2.ms	834
out.loam.2.infilD	200, 0, 0, 0, 0, 0
out.loam.2.drainD	0.009, 5.784, 19.04, 12.94, 9.454, 7.397
out.loam.2.store	370.1, 364.3, 345.2, 332.3, 322.8, 315.4
out.loam.4.infil	200
out.loam.4.drain	54.58
out.loam.4.runoff	0
out.loam.4.store0	None
out.loam.4.storeEnd	315.5
out.loam.4.maxErr	0
out.loam.4.sumErr	0
out.loam.4.iters	7345
out.loam.4.ms	477
out.loam.4.infilD	200, 0, 0, 0, 0, 0
out.loam.4.drainD	0.009, 5.825, 18.99, 12.88, 9.466, 7.402
out.loam.4.store	370.1, 364.2, 345.2, 332.4, 322.9, 315.5
out.loam.0.5.infil	200
out.loam.0.5.drain	54.62
out.loam.0.5.runoff	0
out.loam.0.5.store0	None
out.loam.0.5.storeEnd	315.4
out.loam.0.5.maxErr	0
out.loam.0.5.sumErr	0
out.loam.0.5.iters	14781
out.loam.0.5.ms	4257
out.loam.0.5.infilD	200, 0, 0, 0, 0, 0
out.loam.0.5.drainD	0.009, 5.773, 19.09, 12.9, 9.463, 7.389
out.loam.0.5.store	370.1, 364.3, 345.2, 332.3, 322.8, 315.4
out.siltclayloam.1.infil	21.77
out.siltclayloam.1.drain	0.156
out.siltclayloam.1.runoff	178.2
out.siltclayloam.1.store0	None
out.siltclayloam.1.storeEnd	416.4
out.siltclayloam.1.maxErr	0.05
out.siltclayloam.1.sumErr	0.05
out.siltclayloam.1.iters	1345673
out.siltclayloam.1.ms	182464
out.siltclayloam.1.infilD	21.77, 0, 0, 0, 0, 0

quantity	value
out.siltclayloam.1.drainD	0.026, 0.026, 0.026, 0.026, 0.026, 0.026
out.siltclayloam.1.store	416.6, 416.6, 416.5, 416.5, 416.5, 416.4
out.siltclayloam.2.infil	22.54
out.siltclayloam.2.drain	0.156
out.siltclayloam.2.runoff	177.5
out.siltclayloam.2.store0	None
out.siltclayloam.2.storeEnd	417.2
out.siltclayloam.2.maxErr	0.074
out.siltclayloam.2.sumErr	0.074
out.siltclayloam.2.iters	266393
out.siltclayloam.2.ms	20684
out.siltclayloam.2.infilD	22.54, 0, 0, 0, 0, 0
out.siltclayloam.2.drainD	0.026, 0.026, 0.026, 0.026, 0.026, 0.026
out.siltclayloam.2.store	417.4, 417.3, 417.3, 417.3, 417.3, 417.2
out.siltclayloam.4.infil	24.38
out.siltclayloam.4.drain	0.156
out.siltclayloam.4.runoff	175.6
out.siltclayloam.4.store0	None
out.siltclayloam.4.storeEnd	419
out.siltclayloam.4.maxErr	0.007
out.siltclayloam.4.sumErr	-0.007
out.siltclayloam.4.iters	385896
out.siltclayloam.4.ms	21266
out.siltclayloam.4.infilD	24.38, 0, 0, 0, 0, 0
out.siltclayloam.4.drainD	0.026, 0.026, 0.026, 0.026, 0.026, 0.026
out.siltclayloam.4.store	419.1, 419.1, 419.1, 419, 419, 419

out_p4_richards_ref.json

Source: `Manuscripts/reproducibility/p456/out_p4_richards_ref.json`; 8 values.

quantity	value
sand.infil	200
sand.drain	152.6
sand.storeEnd	93.12
sand.S0	45.68
loam.infil	200
loam.drain	55.12
loam.storeEnd	314.9
loam.S0	170.1

out_p4_runtime.csv

Source: `Manuscripts/reproducibility/p456/out_p4_runtime.csv`; 17 rows. Workbook sheet (`CSV file`).

exp	name	study	study_hours	gui	gui_hours
E0	Baselines	10	0.6411	9	0.5925
E1	Dose and duration	200	10.99	17	0.8878
E2	Trigger pressure	200	7.731	21	0.805
E3	Sensor depth	200	8.099	21	0.84
E4	Methods	100	4.1	6	0.2408
E5	Soil structure dynamics (SHP)	300	15.77	72	3.45
E6	Mediterranean pollution	210	6.937	30	0.7833
E7	Mulching	200	8.949	nan	nan
E8	Rain intensity and moisture	1000	1.727	nan	nan
E9	Soil moisture and rain	1000	1.957	40	0.07778
E10	Amount, timing and soil	1000	70.39	20	0.7111
E11	Temperature, evaporation, rain	1000	41.98	16	0.6578
E12	Sensor, trigger, rain	1000	42.92	30	1.225
E13	N, water, soil	1000	57.44	32	1.929
E14	Mulch and rain	1000	48.01	nan	nan
E15	Method, depth, amount	1000	43.58	30	1.275
E16	Edge cases	770	48.22	24	1.21

out_p5_bridge.json

Source: `Manuscripts/reproducibility/p456/out_p5_bridge.json`; 20 values.

quantity	value
parser[0]	annual-field-nitrogen-3b.dlf, 11, 11, 2.246e-16
parser[1]	annual-field-nitrogen-4b.dlf, 11, 11, 1.703e-16
parser[2]	annual-tracer-3b.dlf, 11, 11, 7.006e-34
parser[3]	daily-water-flux.dlf, 26, 26, 3.277e-16
parser[4]	optim-field-nitrogen.dlf, 1, 1, 0
parser[5]	soil-chemical-110cm.dlf, 11665, 11665, 2.918e-15
n.annual-field-nitrogen-3b.dlf.Nfert	0
n.annual-field-nitrogen-3b.dlf.Nup	0.045
n.annual-field-nitrogen-3b.dlf.NL	0.00478
n.annual-field-nitrogen-3b.dlf.Nmin	0.049
n.annual-field-nitrogen-3b.dlf.Nden	0.00473
n.annual-field-nitrogen-3b.dlf.maxabs_Error_kgN	0
n.annual-field-nitrogen-3b.dlf.seasons	10
n.annual-field-nitrogen-4b.dlf.Nfert	0
n.annual-field-nitrogen-4b.dlf.Nup	0.049
n.annual-field-nitrogen-4b.dlf.NL	0.0041
n.annual-field-nitrogen-4b.dlf.Nmin	0.048

quantity	value
n.annual-field-nitrogen-4b.dlf.Nden	0.00442
n.annual-field-nitrogen-4b.dlf.maxabs_Error_kgN	0
n.annual-field-nitrogen-4b.dlf.seasons	10

out_p5_bridge_js.json

Source: `Manuscripts/reproducibility/p456/out_p5_bridge_js.json`; 832 values, first 220 shown.

quantity	value
deniArr[0]	3.206e-11, 3.206e-11, 1.88e-11, 4.709e-11, 4.709e-11, 2.869e-11
deniArr[1]	3.83e-11, 3.83e-11, 2.235e-11, 5.479e-11, 5.479e-11, 3.35e-11
deniArr[2]	4.481e-11, 4.481e-11, 2.599e-11, 6.21e-11, 6.21e-11, 3.816e-11
deniArr[3]	3.869e-11, 3.869e-11, 2.239e-11, 5.305e-11, 5.305e-11, 3.265e-11
deniArr[4]	2.408e-11, 2.408e-11, 1.396e-11, 3.312e-11, 3.312e-11, 2.037e-11
deniArr[5]	5.238e-12, 5.238e-12, 3.111e-12, 1.273e-11, 1.273e-11, 7.671e-12
deniArr[6]	2.994e-12, 2.994e-12, 1.785e-12, 7.394e-12, 7.394e-12, 4.447e-12
deniArr[7]	1.63e-12, 1.63e-12, 9.747e-13, 4.065e-12, 4.065e-12, 2.442e-12
deniArr[8]	8.865e-13, 8.865e-13, 5.309e-13, 2.222e-12, 2.222e-12, 1.334e-12
deniArr[9]	4.67e-13, 4.67e-13, 2.799e-13, 1.174e-12, 1.174e-12, 7.048e-13
deniArr[10]	2.463e-13, 2.463e-13, 1.477e-13, 6.204e-13, 6.204e-13, 3.723e-13
deniArr[11]	1.221e-13, 1.221e-13, 7.327e-14, 3.079e-13, 3.079e-13, 1.848e-13
deniArr[12]	2.007e-13, 2.007e-13, 1.204e-13, 2.665e-13, 2.665e-13, 1.599e-13
deniArr[13]	9.756e-14, 9.756e-14, 5.853e-14, 1.295e-13, 1.295e-13, 7.772e-14
deniArr[14]	0, 0, 0, 0, 0, 0
deniArr[15]	0, 0, 0, 0, 0, 0
deniArr[16]	0, 0, 0, 0, 0, 0
deniArr[17]	0, 0, 0, 0, 0, 0
deniArr[18]	0, 0, 0, 0, 0, 0
deniArr[19]	0, 0, 0, 0, 0, 0
deniArr[20]	0, 0, 0, 0, 0, 0
deniArr[21]	0, 0, 0, 0, 0, 0
deniArr[22]	0, 0, 0, 0, 0, 0
deniArr[23]	0, 0, 0, 0, 0, 0
deniArr[24]	0, 0, 0, 0, 0, 0
deniArr[25]	0, 0, 0, 0, 0, 0
deniArr[26]	0, 0, 0, 0, 0, 0
deniArr[27]	0, 0, 0, 0, 0, 0
deniArr[28]	0, 0, 0, 0, 0, 0
deniArr[29]	0, 0, 0, 0, 0, 0
deniArr[30]	0, 0, 0, 0, 0, 0
deniArr[31]	0, 0, 0, 0, 0, 0
deniArr[32]	0, 0, 0, 0, 0, 0
deniArr[33]	1.124e-10, 1.124e-10, 6.329e-11, 1.363e-10, 1.363e-10, 8.598e-11
deniArr[34]	7.183e-11, 7.183e-11, 4.113e-11, 9.431e-11, 9.431e-11, 5.856e-11
deniArr[35]	5.099e-11, 5.099e-11, 2.946e-11, 6.949e-11, 6.949e-11, 4.283e-11
deniArr[36]	3.823e-11, 3.823e-11, 2.213e-11, 5.249e-11, 5.249e-11, 3.23e-11

quantity	value
deniArr[37]	2.373e-11, 2.373e-11, 1.376e-11, 3.264e-11, 3.264e-11, 2.006e-11
deniArr[38]	5.254e-12, 5.254e-12, 3.12e-12, 1.276e-11, 1.276e-11, 7.687e-12
deniArr[39]	2.973e-12, 2.973e-12, 1.773e-12, 7.343e-12, 7.343e-12, 4.417e-12
deniArr[40]	1.634e-12, 1.634e-12, 9.773e-13, 4.075e-12, 4.075e-12, 2.448e-12
deniArr[41]	8.821e-13, 8.821e-13, 5.283e-13, 2.211e-12, 2.211e-12, 1.328e-12
deniArr[42]	4.679e-13, 4.679e-13, 2.805e-13, 1.177e-12, 1.177e-12, 7.062e-13
deniArr[43]	2.457e-13, 2.457e-13, 1.474e-13, 6.189e-13, 6.189e-13, 3.714e-13
deniArr[44]	1.221e-13, 1.221e-13, 7.323e-14, 3.079e-13, 3.079e-13, 1.848e-13
deniArr[45]	2.007e-13, 2.007e-13, 1.204e-13, 2.666e-13, 2.666e-13, 1.6e-13
deniArr[46]	9.747e-14, 9.747e-14, 5.848e-14, 1.294e-13, 1.294e-13, 7.765e-14
deniArr[47]	0, 0, 0, 0, 0, 0
deniArr[48]	0, 0, 0, 0, 0, 0
deniArr[49]	0, 0, 0, 0, 0, 0
deniArr[50]	0, 0, 0, 0, 0, 0
deniArr[51]	0, 0, 0, 0, 0, 0
deniArr[52]	0, 0, 0, 0, 0, 0
deniArr[53]	0, 0, 0, 0, 0, 0
deniArr[54]	0, 0, 0, 0, 0, 0
deniArr[55]	0, 0, 0, 0, 0, 0
deniArr[56]	0, 0, 0, 0, 0, 0
deniArr[57]	0, 0, 0, 0, 0, 0
deniArr[58]	0, 0, 0, 0, 0, 0
deniArr[59]	0, 0, 0, 0, 0, 0
deniArr[60]	0, 0, 0, 0, 0, 0
deniArr[61]	0, 0, 0, 0, 0, 0
deniArr[62]	0, 0, 0, 0, 0, 0
deniArr[63]	0, 0, 0, 0, 0, 0
deniArr[64]	0, 0, 0, 0, 0, 0
deniArr[65]	0, 0, 0, 0, 0, 0
deniArr[66]	1.964e-10, 1.964e-10, 1.074e-10, 2.027e-10, 2.027e-10, 1.32e-10
deniArr[67]	1.222e-10, 1.222e-10, 6.842e-11, 1.432e-10, 1.432e-10, 9.084e-11
deniArr[68]	6.988e-11, 6.988e-11, 3.993e-11, 9.051e-11, 9.051e-11, 5.63e-11
deniArr[69]	4.024e-11, 4.024e-11, 2.326e-11, 5.486e-11, 5.486e-11, 3.379e-11
deniArr[70]	2.371e-11, 2.371e-11, 1.376e-11, 3.255e-11, 3.255e-11, 2e-11
deniArr[71]	5.244e-12, 5.244e-12, 3.114e-12, 1.272e-11, 1.272e-11, 7.665e-12
deniArr[72]	2.963e-12, 2.963e-12, 1.767e-12, 7.321e-12, 7.321e-12, 4.403e-12
deniArr[73]	1.632e-12, 1.632e-12, 9.761e-13, 4.07e-12, 4.07e-12, 2.445e-12
deniArr[74]	8.806e-13, 8.806e-13, 5.274e-13, 2.208e-12, 2.208e-12, 1.325e-12
deniArr[75]	4.677e-13, 4.677e-13, 2.803e-13, 1.176e-12, 1.176e-12, 7.058e-13
deniArr[76]	2.455e-13, 2.455e-13, 1.472e-13, 6.183e-13, 6.183e-13, 3.711e-13
deniArr[77]	1.22e-13, 1.22e-13, 7.318e-14, 3.077e-13, 3.077e-13, 1.847e-13
deniArr[78]	2.006e-13, 2.006e-13, 1.204e-13, 2.666e-13, 2.666e-13, 1.6e-13
deniArr[79]	9.741e-14, 9.741e-14, 5.844e-14, 1.293e-13, 1.293e-13, 7.76e-14
deniArr[80]	0, 0, 0, 0, 0, 0
deniArr[81]	0, 0, 0, 0, 0, 0
deniArr[82]	0, 0, 0, 0, 0, 0
deniArr[83]	0, 0, 0, 0, 0, 0
deniArr[84]	0, 0, 0, 0, 0, 0

quantity	value
deniArr[85]	0, 0, 0, 0, 0, 0
deniArr[86]	0, 0, 0, 0, 0, 0
deniArr[87]	0, 0, 0, 0, 0, 0
deniArr[88]	0, 0, 0, 0, 0, 0
deniArr[89]	0, 0, 0, 0, 0, 0
deniArr[90]	0, 0, 0, 0, 0, 0
deniArr[91]	0, 0, 0, 0, 0, 0
deniArr[92]	0, 0, 0, 0, 0, 0
deniArr[93]	0, 0, 0, 0, 0, 0
deniArr[94]	0, 0, 0, 0, 0, 0
deniArr[95]	0, 0, 0, 0, 0, 0
deniArr[96]	0, 0, 0, 0, 0, 0
deniArr[97]	0, 0, 0, 0, 0, 0
deniArr[98]	0, 0, 0, 0, 0, 0
deniArr[99]	2.86e-10, 2.86e-10, 1.53e-10, 2.548e-10, 2.548e-10, 1.715e-10
deniArr[100]	1.737e-10, 1.737e-10, 9.543e-11, 1.836e-10, 1.836e-10, 1.189e-10
deniArr[101]	9.365e-11, 9.365e-11, 5.287e-11, 1.142e-10, 1.142e-10, 7.186e-11
deniArr[102]	4.498e-11, 4.498e-11, 2.591e-11, 6.029e-11, 6.029e-11, 3.725e-11
deniArr[103]	2.415e-11, 2.415e-11, 1.4e-11, 3.3e-11, 3.3e-11, 2.029e-11
deniArr[104]	5.245e-12, 5.245e-12, 3.115e-12, 1.271e-11, 1.271e-11, 7.659e-12
deniArr[105]	2.956e-12, 2.956e-12, 1.763e-12, 7.303e-12, 7.303e-12, 4.392e-12
deniArr[106]	1.629e-12, 1.629e-12, 9.744e-13, 4.063e-12, 4.063e-12, 2.441e-12
deniArr[107]	8.794e-13, 8.794e-13, 5.267e-13, 2.205e-12, 2.205e-12, 1.324e-12
deniArr[108]	4.673e-13, 4.673e-13, 2.801e-13, 1.175e-12, 1.175e-12, 7.052e-13
deniArr[109]	2.454e-13, 2.454e-13, 1.471e-13, 6.179e-13, 6.179e-13, 3.708e-13
deniArr[110]	1.219e-13, 1.219e-13, 7.312e-14, 3.076e-13, 3.076e-13, 1.846e-13
deniArr[111]	2.004e-13, 2.004e-13, 1.203e-13, 2.664e-13, 2.664e-13, 1.599e-13
deniArr[112]	9.735e-14, 9.735e-14, 5.841e-14, 1.293e-13, 1.293e-13, 7.756e-14
deniArr[113]	0, 0, 0, 0, 0, 0
deniArr[114]	0, 0, 0, 0, 0, 0
deniArr[115]	0, 0, 0, 0, 0, 0
deniArr[116]	0, 0, 0, 0, 0, 0
deniArr[117]	0, 0, 0, 0, 0, 0
deniArr[118]	0, 0, 0, 0, 0, 0
deniArr[119]	0, 0, 0, 0, 0, 0
deniArr[120]	0, 0, 0, 0, 0, 0
deniArr[121]	0, 0, 0, 0, 0, 0
deniArr[122]	0, 0, 0, 0, 0, 0
deniArr[123]	0, 0, 0, 0, 0, 0
deniArr[124]	0, 0, 0, 0, 0, 0
deniArr[125]	0, 0, 0, 0, 0, 0
deniArr[126]	0, 0, 0, 0, 0, 0
deniArr[127]	0, 0, 0, 0, 0, 0
deniArr[128]	0, 0, 0, 0, 0, 0
deniArr[129]	0, 0, 0, 0, 0, 0
deniArr[130]	0, 0, 0, 0, 0, 0
deniArr[131]	0, 0, 0, 0, 0, 0
deniArr[132]	3.764e-10, 3.764e-10, 1.979e-10, 2.94e-10, 2.94e-10, 2.043e-10

quantity	value
deniArr[133]	2.296e-10, 2.296e-10, 1.242e-10, 2.196e-10, 2.196e-10, 1.453e-10
deniArr[134]	1.196e-10, 1.196e-10, 6.675e-11, 1.373e-10, 1.373e-10, 8.741e-11
deniArr[135]	5.156e-11, 5.156e-11, 2.956e-11, 6.756e-11, 6.756e-11, 4.191e-11
deniArr[136]	2.508e-11, 2.508e-11, 1.453e-11, 3.405e-11, 3.405e-11, 2.095e-11
deniArr[137]	5.274e-12, 5.274e-12, 3.132e-12, 1.277e-11, 1.277e-11, 7.693e-12
deniArr[138]	2.952e-12, 2.952e-12, 1.761e-12, 7.293e-12, 7.293e-12, 4.386e-12
deniArr[139]	1.627e-12, 1.627e-12, 9.728e-13, 4.057e-12, 4.057e-12, 2.437e-12
deniArr[140]	8.783e-13, 8.783e-13, 5.26e-13, 2.202e-12, 2.202e-12, 1.322e-12
deniArr[141]	4.669e-13, 4.669e-13, 2.799e-13, 1.174e-12, 1.174e-12, 7.046e-13
deniArr[142]	2.452e-13, 2.452e-13, 1.47e-13, 6.175e-13, 6.175e-13, 3.706e-13
deniArr[143]	1.218e-13, 1.218e-13, 7.306e-14, 3.074e-13, 3.074e-13, 1.845e-13
deniArr[144]	2.003e-13, 2.003e-13, 1.201e-13, 2.663e-13, 2.663e-13, 1.598e-13
deniArr[145]	9.729e-14, 9.729e-14, 5.837e-14, 1.292e-13, 1.292e-13, 7.751e-14
deniArr[146]	0, 0, 0, 0, 0, 0
deniArr[147]	0, 0, 0, 0, 0, 0
deniArr[148]	0, 0, 0, 0, 0, 0
deniArr[149]	0, 0, 0, 0, 0, 0
deniArr[150]	0, 0, 0, 0, 0, 0
deniArr[151]	0, 0, 0, 0, 0, 0
deniArr[152]	0, 0, 0, 0, 0, 0
deniArr[153]	0, 0, 0, 0, 0, 0
deniArr[154]	0, 0, 0, 0, 0, 0
deniArr[155]	0, 0, 0, 0, 0, 0
deniArr[156]	0, 0, 0, 0, 0, 0
deniArr[157]	0, 0, 0, 0, 0, 0
deniArr[158]	0, 0, 0, 0, 0, 0
deniArr[159]	0, 0, 0, 0, 0, 0
deniArr[160]	0, 0, 0, 0, 0, 0
deniArr[161]	0, 0, 0, 0, 0, 0
deniArr[162]	0, 0, 0, 0, 0, 0
deniArr[163]	0, 0, 0, 0, 0, 0
deniArr[164]	0, 0, 0, 0, 0, 0
deniArr[165]	4.711e-10, 4.711e-10, 2.444e-10, 3.245e-10, 3.245e-10, 2.33e-10
deniArr[166]	2.868e-10, 2.868e-10, 1.53e-10, 2.497e-10, 2.497e-10, 1.689e-10
deniArr[167]	1.476e-10, 1.476e-10, 8.148e-11, 1.594e-10, 1.594e-10, 1.027e-10
deniArr[168]	5.941e-11, 5.941e-11, 3.389e-11, 7.585e-11, 7.585e-11, 4.727e-11
deniArr[169]	2.648e-11, 2.648e-11, 1.532e-11, 3.562e-11, 3.562e-11, 2.194e-11
deniArr[170]	5.342e-12, 5.342e-12, 3.172e-12, 1.291e-11, 1.291e-11, 7.781e-12
deniArr[171]	2.954e-12, 2.954e-12, 1.762e-12, 7.297e-12, 7.297e-12, 4.389e-12
deniArr[172]	1.625e-12, 1.625e-12, 9.717e-13, 4.052e-12, 4.052e-12, 2.434e-12
deniArr[173]	8.773e-13, 8.773e-13, 5.254e-13, 2.199e-12, 2.199e-12, 1.321e-12
deniArr[174]	4.665e-13, 4.665e-13, 2.796e-13, 1.173e-12, 1.173e-12, 7.04e-13
deniArr[175]	2.45e-13, 2.45e-13, 1.47e-13, 6.171e-13, 6.171e-13, 3.703e-13
deniArr[176]	1.217e-13, 1.217e-13, 7.299e-14, 3.072e-13, 3.072e-13, 1.843e-13
deniArr[177]	2.001e-13, 2.001e-13, 1.2e-13, 2.661e-13, 2.661e-13, 1.597e-13
deniArr[178]	9.722e-14, 9.722e-14, 5.833e-14, 1.291e-13, 1.291e-13, 7.746e-14
deniArr[179]	0, 0, 0, 0, 0, 0
deniArr[180]	0, 0, 0, 0, 0, 0

quantity	value
deniArr[181]	0, 0, 0, 0, 0, 0
deniArr[182]	0, 0, 0, 0, 0, 0
deniArr[183]	0, 0, 0, 0, 0, 0
deniArr[184]	0, 0, 0, 0, 0, 0
deniArr[185]	0, 0, 0, 0, 0, 0
deniArr[186]	0, 0, 0, 0, 0, 0
deniArr[187]	0, 0, 0, 0, 0, 0
deniArr[188]	0, 0, 0, 0, 0, 0
deniArr[189]	0, 0, 0, 0, 0, 0
deniArr[190]	0, 0, 0, 0, 0, 0
deniArr[191]	0, 0, 0, 0, 0, 0
deniArr[192]	0, 0, 0, 0, 0, 0
deniArr[193]	0, 0, 0, 0, 0, 0
deniArr[194]	0, 0, 0, 0, 0, 0
deniArr[195]	0, 0, 0, 0, 0, 0
deniArr[196]	0, 0, 0, 0, 0, 0
deniArr[197]	0, 0, 0, 0, 0, 0
deniArr[198]	5.655e-10, 5.655e-10, 2.901e-10, 3.469e-10, 3.469e-10, 2.573e-10
deniArr[199]	3.463e-10, 3.463e-10, 1.826e-10, 2.755e-10, 2.755e-10, 1.904e-10
deniArr[200]	1.769e-10, 1.769e-10, 9.67e-11, 1.8e-10, 1.8e-10, 1.175e-10
deniArr[201]	6.822e-11, 6.822e-11, 3.87e-11, 8.468e-11, 8.468e-11, 5.304e-11
deniArr[202]	2.827e-11, 2.827e-11, 1.632e-11, 3.762e-11, 3.762e-11, 2.321e-11
deniArr[203]	5.45e-12, 5.45e-12, 3.236e-12, 1.315e-11, 1.315e-11, 7.925e-12
deniArr[204]	2.964e-12, 2.964e-12, 1.768e-12, 7.322e-12, 7.322e-12, 4.404e-12
deniArr[205]	1.624e-12, 1.624e-12, 9.713e-13, 4.05e-12, 4.05e-12, 2.433e-12
deniArr[206]	8.764e-13, 8.764e-13, 5.249e-13, 2.197e-12, 2.197e-12, 1.319e-12
deniArr[207]	4.661e-13, 4.661e-13, 2.794e-13, 1.172e-12, 1.172e-12, 7.034e-13
deniArr[208]	2.449e-13, 2.449e-13, 1.469e-13, 6.167e-13, 6.167e-13, 3.701e-13
deniArr[209]	1.216e-13, 1.216e-13, 7.292e-14, 3.07e-13, 3.07e-13, 1.842e-13
deniArr[210]	1.998e-13, 1.998e-13, 1.199e-13, 2.659e-13, 2.659e-13, 1.596e-13
deniArr[211]	9.715e-14, 9.715e-14, 5.829e-14, 1.29e-13, 1.29e-13, 7.741e-14
deniArr[212]	0, 0, 0, 0, 0, 0
deniArr[213]	0, 0, 0, 0, 0, 0
deniArr[214]	0, 0, 0, 0, 0, 0
deniArr[215]	0, 0, 0, 0, 0, 0
deniArr[216]	0, 0, 0, 0, 0, 0
deniArr[217]	0, 0, 0, 0, 0, 0
deniArr[218]	0, 0, 0, 0, 0, 0
deniArr[219]	0, 0, 0, 0, 0, 0

out_p5_chem.json

Source: `Manuscripts/reproducibility/p456/out_p5_chem.json`; 6 values.

quantity	value
resid	-0.1097, 0.1082, 0.0011, -3.553e-15, 0.0007, -0.1228, 0.1221, 0, -0.0002, 0.001
maxAbs	0.1228
sumAbs	0.4658
applied	2290
leachTotal	393
uptakeTotal	1934

out_p6_crops.json

Source: `Manuscripts/reproducibility/p456/out_p6_crops.json`; 1495 values, first 220 shown.

quantity	value
n	48
groups.field	19
groups.perennial	29
habits.annual	19
habits.deciduous	18
habits.evergreen	11
combos	110
issues[0][0]	citrus
issues[0][1]	Young
issues[0][2]	Kc mid < Kc ini
issues[1][0]	citrus
issues[1][1]	Low density
issues[1][2]	Kc mid < Kc ini
issues[2][0]	citrus
issues[2][1]	Medium density, short
issues[2][2]	Kc mid < Kc ini
issues[3][0]	citrus
issues[3][1]	Medium density, tall
issues[3][2]	Kc mid < Kc ini
issues[4][0]	citrus
issues[4][1]	High density, short
issues[4][2]	Kc mid < Kc ini
issues[5][0]	citrus
issues[5][1]	High density, tall
issues[5][2]	Kc mid < Kc ini
issues[6][0]	olive
issues[6][1]	Young
issues[6][2]	Kc mid < Kc ini, Kc mid 0.25
issues[7][0]	olive
issues[7][1]	Traditional, low density
issues[7][2]	Kc mid < Kc ini, Kc mid 0.3
issues[8][0]	olive
issues[8][1]	Traditional, medium density

quantity	value
issues[8][2]	Kc mid < Kc ini
issues[9][0]	olive
issues[9][1]	Intensive (hedge pruned)
issues[9][2]	Kc mid < Kc ini
issues[10][0]	olive
issues[10][1]	Super-intensive, medium
issues[10][2]	Kc mid < Kc ini
issues[11][0]	olive
issues[11][1]	Super-intensive, high
issues[11][2]	Kc mid < Kc ini
issues[12][0]	datepalm
issues[12][1]	Young, low
issues[12][2]	Kc mid < Kc ini
rows[0].key	rice
rows[0].dens	None
rows[0].group	field
rows[0].habit	annual
rows[0].kcIni	1.05
rows[0].kcMid	1.2
rows[0].kcEnd	0.9
rows[0].len	120
rows[0].root	75
rows[0].p	0.2
rows[0].ky	None
rows[0].kcSum	135.1
rows[0].iss	
rows[1].key	maize
rows[1].dens	None
rows[1].group	field
rows[1].habit	annual
rows[1].kcIni	0.3
rows[1].kcMid	1.2
rows[1].kcEnd	0.6
rows[1].len	105
rows[1].root	135
rows[1].p	0.55
rows[1].ky	1.25
rows[1].kcSum	92.85
rows[1].iss	
rows[2].key	pearlmillet
rows[2].dens	None
rows[2].group	field
rows[2].habit	annual
rows[2].kcIni	0.3
rows[2].kcMid	1
rows[2].kcEnd	0.3
rows[2].len	105
rows[2].root	150

quantity	value
rows[2].p	0.55
rows[2].ky	None
rows[2].kcSum	77
rows[2].iss	
rows[3].key	sorghum
rows[3].dens	None
rows[3].group	field
rows[3].habit	annual
rows[3].kcIni	0.3
rows[3].kcMid	1
rows[3].kcEnd	0.55
rows[3].len	125
rows[3].root	150
rows[3].p	0.55
rows[3].ky	0.9
rows[3].kcSum	91.88
rows[3].iss	
rows[4].key	cassava
rows[4].dens	None
rows[4].group	field
rows[4].habit	annual
rows[4].kcIni	0.3
rows[4].kcMid	0.8
rows[4].kcEnd	0.3
rows[4].len	210
rows[4].root	65
rows[4].p	0.35
rows[4].ky	None
rows[4].kcSum	133
rows[4].iss	
rows[5].key	yam
rows[5].dens	None
rows[5].group	field
rows[5].habit	annual
rows[5].kcIni	0.5
rows[5].kcMid	1.1
rows[5].kcEnd	0.85
rows[5].len	240
rows[5].root	60
rows[5].p	0.4
rows[5].ky	None
rows[5].kcSum	220.3
rows[5].iss	
rows[6].key	cowpea
rows[6].dens	None
rows[6].group	field
rows[6].habit	annual
rows[6].kcIni	0.4

quantity	value
rows[6].kcMid	1.05
rows[6].kcEnd	0.5
rows[6].len	100
rows[6].root	80
rows[6].p	0.45
rows[6].ky	None
rows[6].kcSum	76.7
rows[6].iss	
rows[7].key	greengram
rows[7].dens	None
rows[7].group	field
rows[7].habit	annual
rows[7].kcIni	0.4
rows[7].kcMid	0.96
rows[7].kcEnd	0.51
rows[7].len	90
rows[7].root	75
rows[7].p	0.45
rows[7].ky	None
rows[7].kcSum	66.78
rows[7].iss	
rows[8].key	soybean
rows[8].dens	None
rows[8].group	field
rows[8].habit	annual
rows[8].kcIni	0.4
rows[8].kcMid	1.15
rows[8].kcEnd	0.5
rows[8].len	120
rows[8].root	95
rows[8].p	0.5
rows[8].ky	0.85
rows[8].kcSum	103.6
rows[8].iss	
rows[9].key	groundnut
rows[9].dens	None
rows[9].group	field
rows[9].habit	annual
rows[9].kcIni	0.4
rows[9].kcMid	1.15
rows[9].kcEnd	0.6
rows[9].len	120
rows[9].root	75
rows[9].p	0.5
rows[9].ky	0.7
rows[9].kcSum	102.9
rows[9].iss	
rows[10].key	cotton

quantity	value
rows[10].dens	None
rows[10].group	field
rows[10].habit	annual
rows[10].kcIni	0.35
rows[10].kcMid	1.15
rows[10].kcEnd	0.5
rows[10].len	190
rows[10].root	135
rows[10].p	0.65
rows[10].ky	0.85
rows[10].kcSum	160.2
rows[10].iss	
rows[11].key	wheat
rows[11].dens	None
rows[11].group	field
rows[11].habit	annual
rows[11].kcIni	0.4
rows[11].kcMid	1.15
rows[11].kcEnd	0.41
rows[11].len	120
rows[11].root	165
rows[11].p	0.55
rows[11].ky	1.05
rows[11].kcSum	102.5
rows[11].iss	
rows[12].key	chickpea
rows[12].dens	None
rows[12].group	field
rows[12].habit	annual
rows[12].kcIni	0.4
rows[12].kcMid	1
rows[12].kcEnd	0.4
rows[12].len	120
rows[12].root	80
rows[12].p	0.5
rows[12].ky	None
rows[12].kcSum	93
rows[12].iss	
rows[13].key	mustard
rows[13].dens	None
rows[13].group	field
rows[13].habit	annual
rows[13].kcIni	0.35

out_p6_pheno.json

Source: `Manuscripts/reproducibility/p456/out_p6_pheno.json`; 560 values, first 220 shown.

quantity	value
[0].clim	warm
[0].geno	Reference
[0].tr	ww
[0].closure_mm	-6.724e-07
[0].det	0
[0].P	0
[0].I	589.1
[0].T	330
[0].Es	252.3
[0].drain	0
[0].grain	515.1
[0].shoot	542.3
[0].root	172.6
[0].rdepth	119.5
[0].cwsiMax	0.2467
[0].ctdMax	1.066
[0].psiMin	-1.213
[0].lai	2.632
[0].height	213.7
[0].stage	3.657
[1].clim	warm
[1].geno	Reference
[1].tr	dr
[1].closure_mm	2.013e-05
[1].det	0
[1].P	0
[1].I	519.7
[1].T	253.9
[1].Es	259.1
[1].drain	1.907e-06
[1].grain	345.4
[1].shoot	478.6
[1].root	166.1
[1].rdepth	119.5
[1].cwsiMax	0.3367
[1].ctdMax	1.076
[1].psiMin	-1.285
[1].lai	2.429
[1].height	213
[1].stage	3.657
[2].clim	warm
[2].geno	Deep rooting
[2].tr	ww

quantity	value
[2].closure_mm	-2.057e-05
[2].det	0
[2].P	0
[2].I	584.2
[2].T	305.8
[2].Es	271.6
[2].drain	0
[2].grain	485.8
[2].shoot	481.6
[2].root	186.1
[2].rdepth	119.5
[2].cwsiMax	0.2172
[2].ctdMax	1.078
[2].psiMin	-1.203
[2].lai	2.326
[2].height	213.7
[2].stage	3.657
[3].clim	warm
[3].geno	Deep rooting
[3].tr	dr
[3].closure_mm	-3.866e-05
[3].det	0
[3].P	0
[3].I	518.4
[3].T	239.2
[3].Es	272.4
[3].drain	0
[3].grain	328.9
[3].shoot	432.8
[3].root	179.2
[3].rdepth	119.5
[3].cwsiMax	0.346
[3].ctdMax	1.085
[3].psiMin	-1.282
[3].lai	2.192
[3].height	213.1
[3].stage	3.657
[4].clim	warm
[4].geno	Shallow, dense
[4].tr	ww
[4].closure_mm	-1.514e-05
[4].det	0
[4].P	0
[4].I	589.6
[4].T	331.3
[4].Es	251.6
[4].drain	0
[4].grain	516.2

quantity	value
[4].shoot	544
[4].root	172.7
[4].rdepth	109
[4].cwsiMax	0.2477
[4].ctdMax	1.066
[4].psiMin	-1.214
[4].lai	2.644
[4].height	213.6
[4].stage	3.657
[5].clim	warm
[5].geno	Shallow, dense
[5].tr	dr
[5].closure_mm	2.381e-05
[5].det	0
[5].P	0
[5].I	501.3
[5].T	228.3
[5].Es	266.3
[5].drain	1.907e-06
[5].grain	297.7
[5].shoot	429.1
[5].root	158.4
[5].rdepth	111.5
[5].cwsiMax	0.3839
[5].ctdMax	1.082
[5].psiMin	-1.27
[5].lai	2.219
[5].height	212.7
[5].stage	3.657
[6].clim	warm
[6].geno	Stay-green
[6].tr	ww
[6].closure_mm	1.436e-05
[6].det	0
[6].P	0
[6].I	611.2
[6].T	395
[6].Es	209.6
[6].drain	0
[6].grain	541.7
[6].shoot	552.9
[6].root	176.6
[6].rdepth	119.5
[6].cwsiMax	0.2492
[6].ctdMax	1.113
[6].psiMin	-1.211
[6].lai	2.738
[6].height	213.7

quantity	value
[6].stage	3.657
[7].clim	warm
[7].geno	Stay-green
[7].tr	dr
[7].closure_mm	4.894e-06
[7].det	0
[7].P	0
[7].I	543.4
[7].T	319
[7].Es	217.8
[7].drain	0
[7].grain	376.9
[7].shoot	488.7
[7].root	170.7
[7].rdepth	119.5
[7].cwsiMax	0.3559
[7].ctdMax	1.11
[7].psiMin	-1.265
[7].lai	2.576
[7].height	213
[7].stage	3.657
[8].clim	warm
[8].geno	Water saver
[8].tr	ww
[8].closure_mm	-1.134e-05
[8].det	0
[8].P	0
[8].I	506.2
[8].T	164.5
[8].Es	334.8
[8].drain	0
[8].grain	330.9
[8].shoot	336
[8].root	143.3
[8].rdepth	119.5
[8].cwsiMax	0.3778
[8].ctdMax	0.9156
[8].psiMin	-0.8846
[8].lai	1.48
[8].height	213.7
[8].stage	3.657
[9].clim	warm
[9].geno	Water saver
[9].tr	dr
[9].closure_mm	-6.134e-06
[9].det	0
[9].P	0
[9].I	453.8

quantity	value
[9].T	135
[9].Es	311.9
[9].drain	0
[9].grain	241.9
[9].shoot	300.1
[9].root	135.6
[9].rdepth	119.5
[9].cwsiMax	0.4086
[9].ctdMax	0.9363
[9].psiMin	-0.9405
[9].lai	1.364
[9].height	213.2
[9].stage	3.657
[10].clim	warm
[10].geno	Water spender
[10].tr	ww
[10].closure_mm	1.024e-05
[10].det	0
[10].P	0
[10].I	615.1
[10].T	377
[10].Es	231.5
[10].drain	0
[10].grain	561.5
[10].shoot	608
[10].root	180.6
[10].rdepth	119.5
[10].cwsiMax	0.1435
[10].ctdMax	1.114
[10].psiMin	-1.59
[10].lai	3.01
[10].height	213.5
[10].stage	3.657

out_p6_sweep.json

Source: `Manuscripts/reproducibility/p456/out_p6_sweep.json`; 286 values, first 220 shown.

quantity	value
[0].id	sid
[0].ran	False
[0].engineInputs	19
[0].ms	6655
[0].tabs	Results, Design, Charts, Figure, Messages
[0].inputs	0
[0].chartsTab	8

quantity	value
[0].errors	
[0].kpi	62
[0].msgs	0
[0].charts	4
[0].textLen	2259
[0].canvas	None
[1].id	drip
[1].ran	False
[1].engineInputs	19
[1].ms	6639
[1].tabs	Results, Design, Charts, Figure, Messages
[1].inputs	0
[1].chartsTab	8
[1].errors	
[1].kpi	62
[1].msgs	2
[1].charts	4
[1].textLen	2500
[1].canvas	None
[2].id	spr
[2].ran	False
[2].engineInputs	19
[2].ms	6839
[2].tabs	Results, Design, Charts, Figure, Messages
[2].inputs	0
[2].chartsTab	8
[2].errors	
[2].kpi	62
[2].msgs	4
[2].charts	4
[2].textLen	2860
[2].canvas	None
[3].id	v1
[3].ran	True
[3].engineInputs	19
[3].ms	6431
[3].tabs	Results, Design, Charts, Figure, Messages
[3].inputs	0
[3].chartsTab	4
[3].errors	
[3].kpi	62
[3].msgs	4
[3].charts	4
[3].textLen	1672
[3].canvas	None
[4].id	n1
[4].ran	True
[4].engineInputs	19

quantity	value
[4].ms	6350
[4].tabs	Results, Design, Charts, Figure, Messages
[4].inputs	0
[4].chartsTab	4
[4].errors	
[4].kpi	62
[4].msgs	6
[4].charts	6
[4].textLen	3419
[4].canvas	606, 460
[5].id	v4
[5].ran	True
[5].engineInputs	19
[5].ms	6388
[5].tabs	Results, Design, Charts, Figure, Messages
[5].inputs	0
[5].chartsTab	4
[5].errors	
[5].kpi	62
[5].msgs	4
[5].charts	4
[5].textLen	1584
[5].canvas	None
[6].id	v6
[6].ran	True
[6].engineInputs	19
[6].ms	6371
[6].tabs	Results, Design, Charts, Figure, Messages
[6].inputs	0
[6].chartsTab	4
[6].errors	
[6].kpi	62
[6].msgs	2
[6].charts	4
[6].textLen	1503
[6].canvas	None
[7].id	e1
[7].ran	True
[7].engineInputs	19
[7].ms	6574
[7].tabs	Results, Design, Charts, Figure, Messages
[7].inputs	0
[7].chartsTab	4
[7].errors	
[7].kpi	62
[7].msgs	3
[7].charts	4
[7].textLen	1802

quantity	value
[7].canvas	None
[8].id	n2
[8].ran	True
[8].engineInputs	19
[8].ms	6405
[8].tabs	Results, Design, Charts, Figure, Messages
[8].inputs	0
[8].chartsTab	4
[8].errors	
[8].kpi	62
[8].msgs	2
[8].charts	4
[8].textLen	1674
[8].canvas	None
[9].id	n3
[9].ran	True
[9].engineInputs	19
[9].ms	6370
[9].tabs	Results, Design, Charts, Figure, Messages
[9].inputs	0
[9].chartsTab	4
[9].errors	
[9].kpi	62
[9].msgs	5
[9].charts	4
[9].textLen	1680
[9].canvas	None
[10].id	v3
[10].ran	True
[10].engineInputs	19
[10].ms	6457
[10].tabs	Results, Design, Charts, Figure, Messages
[10].inputs	0
[10].chartsTab	4
[10].errors	
[10].kpi	62
[10].msgs	3
[10].charts	4
[10].textLen	1661
[10].canvas	None
[11].id	v7
[11].ran	True
[11].engineInputs	19
[11].ms	6461
[11].tabs	Results, Design, Charts, Figure, Messages
[11].inputs	0
[11].chartsTab	4
[11].errors	

quantity	value
[11].kpi	62
[11].msgs	0
[11].charts	6
[11].textLen	3237
[11].canvas	606, 440
[12].id	r1
[12].ran	True
[12].engineInputs	19
[12].ms	6429
[12].tabs	Results, Design, Charts, Figure, Messages
[12].inputs	0
[12].chartsTab	4
[12].errors	
[12].kpi	62
[12].msgs	3
[12].charts	4
[12].textLen	1987
[12].canvas	None
[13].id	dph
[13].ran	False
[13].engineInputs	19
[13].ms	6328
[13].tabs	Results, Design, Charts, Figure, Messages
[13].inputs	0
[13].chartsTab	0
[13].errors	
[13].kpi	0
[13].msgs	0
[13].charts	0
[13].textLen	292
[13].canvas	None
[14].id	field
[14].ran	True
[14].engineInputs	19
[14].ms	6716
[14].tabs	Results, Design, Charts, Figure, Messages
[14].inputs	0
[14].chartsTab	4
[14].errors	
[14].kpi	62
[14].msgs	5
[14].charts	4
[14].textLen	2084
[14].canvas	None
[15].id	spac
[15].ran	True
[15].engineInputs	19
[15].ms	6562

quantity	value
[15].tabs	Results, Design, Charts, Figure, Messages
[15].inputs	0
[15].chartsTab	4
[15].errors	
[15].kpi	62
[15].msgs	0
[15].charts	4
[15].textLen	1470
[15].canvas	None
[16].id	v2
[16].ran	True
[16].engineInputs	19
[16].ms	6490
[16].tabs	Results, Design, Charts, Figure, Messages
[16].inputs	0
[16].chartsTab	4
[16].errors	
[16].kpi	62
[16].msgs	4
[16].charts	4
[16].textLen	1695

out_p7_planner.json

Source: `Manuscripts/reproducibility/p456/out_p7_planner.json`; 57 values.

quantity	value
n_questions	8
passed	8
rows[0].q	How does the irrigation trigger of -100 to -800 cm change drainage on a sandy loam in Bologna over 10 years?
rows[0].expected	trig
rows[0].got	irrig, trig
rows[0].station	it-bologna
rows[0].soil	usda:sandyloam
rows[0].pass	True
rows[1].q	What happens to nitrate leaching if I cut the nitrogen dose by 20 % in Foulum?
rows[1].expected	nfac
rows[1].got	nfac, mm
rows[1].station	dk-foulum
rows[1].soil	None
rows[1].pass	True
rows[2].q	Compare 20 % less rain and 20 % more rain at Jyndevad on a loamy sand
rows[2].expected	precip
rows[2].got	precip
rows[2].station	dk-jyndevad

quantity	value
rows[2].soil	usda:loamysand
rows[2].pass	True
rows[3].q	Warming of 2 degrees and 4 degrees in Poland, Jadwisin
rows[3].expected	temp
rows[3].got	temp
rows[3].station	pl-jadwisin
rows[3].soil	None
rows[3].pass	True
rows[4].q	Is drip irrigation better than rainfed on a silt loam in the Czech Republic?
rows[4].expected	irrig
rows[4].got	irrig, method
rows[4].station	cz-valecov
rows[4].soil	usda:siltloam
rows[4].pass	True
rows[5].q	Effect of the sensor depth 10, 30 and 60 cm on the irrigation trigger
rows[5].expected	z
rows[5].got	irrig, trig, z
rows[5].station	None
rows[5].soil	None
rows[5].pass	True
rows[6].q	Pesticide half-life of 20 and 60 days and sorption koc of 50 and 200 l/kg
rows[6].expected	half, koc
rows[6].got	koc, half
rows[6].station	None
rows[6].soil	None
rows[6].pass	True
rows[7].q	What if the rooting depth is 60 and 120 cm in Bratislava?
rows[7].expected	root
rows[7].got	root
rows[7].station	sk-bratislava
rows[7].soil	None
rows[7].pass	True
catalogue_factors	54
catalogue_soils	10
catalogue_stations	8
validate.factors[0].id	precip
validate.factors[0].levels	0.4, 1, 1.6
validate.responses	yield
validate.notes	unknown factor ”not/_a/_factor” left out

S7. Where every table and figure comes from

Item	How it is made	Script(s)	Input and output
Figure 1	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_architecture.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p6/fig01_system.pdf
Table 1	Descriptive table (definitions, design, parameters or tests); no computation	-	
Figure 2	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/zaria_ndvi.jpg; Manuscripts/figures/shots_jpg/zaria_scenes.jpg; Manuscripts/figures/shots_jpg/real_sketch.jpg
Figure 3	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/zaria_terrain.jpg; Manuscripts/figures/shots_jpg/zaria_gndvi_water.jpg; Manuscripts/figures/shots_jpg/zaria_rusle.jpg; Manuscripts/figures/shots_jpg/zaria_available_water.jpg
Figure 4	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/zaria_vri_depth.jpg; Manuscripts/figures/shots_jpg/zaria_matric_potential.jpg
Figure 5	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p6/fig03_pheno_figures.jpg
Figure 6	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/real_season_film.jpg
Figure 7	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_architecture.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p4/fig01_workbench.pdf
Figure 8	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p4/fig02a_process_map.png
Figure 9	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p4/fig02b_soil_column.png
Figure 10	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/wb_step1_input.jpg; Manuscripts/figures/shots_jpg/wb_step2_library.jpg; Manuscripts/figures/shots_jpg/wb_step4_scenarios.jpg; Manuscripts/figures/shots_jpg/wb_step6_logplots.jpg; Manuscripts/figures/shots_jpg/wb_step9_layers.jpg; Manuscripts/figures/shots_jpg/wb_tool_setup_editor.jpg; Manuscripts/figures/shots_jpg/wb_tool_calibrate.jpg; Manuscripts/figures/shots_jpg/wb_tool_learn_files.jpg
Figure 11	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/zaria_main_feedback.jpg
Figure 12	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_schematics.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p5/fig01_bridge.pdf
Table 2	Descriptive table (definitions, design, parameters or tests); no computation	-	
Table 3	Table written by the script (tex/data_table.tex)	Manuscripts/reproducibility/analysis/data_overview.py	Data/Comprehensive_Summary/00_summary_all_simulations
Figure 13	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/data_overview.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/data/fig_data_overview.pdf
Table 4	Table typed from the result file (13 of its numbers found there)	Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json

Item	How it is made	Script(s)	Input and output
Table 5	Table typed in the manuscript from the result files of the script	p4_design_sizes.js; out_p4_design.js	Data/daisy_ensemble.csv.gz (10 190 runs)
Figure 14	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py; p4_design_fidelity.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p4/fig03_fidelity.pdf
Table 6	Table typed from the result file (5 of its numbers found there)	Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py; p4_richards.js; p4_richards_ref.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json
Figure 15	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py; p4_richards.js; p4_richards_ref.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p4/fig04_richards.pdf
Figure 16	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py; p5_bridge_checks.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p4/fig05_denitrification.pdf
Table 7	Descriptive table (definitions, design, parameters or tests); no computation	-	
Table 8	Table typed in the manuscript from the result files of the script	(see the reproducibility appendix of the paper)	Data/daisy_ensemble.csv.gz (10 190 runs)
Table 9	Descriptive table (definitions, design, parameters or tests); no computation	-	
Figure 17	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p5/fig04_economics.pdf
Figure 18	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p5/fig02_flowfigures.jpg
Figure 19	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/ui_layers_tree.jpg; Manuscripts/figures/shots_jpg/ui_field_panel1.jpg; Manuscripts/figures/shots_jpg/ui_field_panel2.jpg
Figure 20	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/wb_summary_effects.jpg; Manuscripts/figures/shots_jpg/wb_every_log_lai.jpg
Figure 21	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/wb_why_failed.jpg; Manuscripts/figures/shots_jpg/wb_all_outputs.jpg
Table 10	Table typed in the manuscript from the result files of the script	(see the reproducibility appendix of the paper)	Data/daisy_ensemble.csv.gz (10 190 runs)
Figure 22	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p6/fig04_pheno.pdf
Figure 23	Figure drawn by the script from the input data	Manuscripts/reproducibility/p456/p456_figs_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p6/fig05_crops.pdf

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